The Dickson torus tower, and a deeper-2-adic Kummer control

research/pkm_dickson_torus_20261015 · 2026-10-15 · lane dickson-torus

Protocol committed before any measured cell: `9776493c4` (PROTOCOL.md sha256 `43e0071a…fb39fb`)

What this is. A stage diagnostic of the solver axis (AGENTS.md §2). It measures the F4 solving degree D of PKM tower systems with `f4_fp_tower`. It prices no pipeline, reports no S and claims no speed. Labels: *measured* is a committed row; *derived* is a calculation or argument; *recorded* is a value from an earlier round (PKM note §10–§16). The source of truth is README.md in the study directory.

1. Answers

question	answer	status
Q-A: does the torus Dickson tower ($2^t \mid p+1$) give the split D?	**H_same.** At both torus primes, torus D = split D in all 16 cells: $m=2$, $N=2..20$; $m=3$, $N=6,9,12,15$ (one target per prime at 15); $m=4$, $N=8,12$. Width, step count and (at the 31-bit primes) multiply-adds also equal the split Dickson's, so it is the same F4 computation on different numbers.	measured
Q-A∞: degenerate at infinity?	**Yes.** The equations are the split Dickson ones with another constant c, so the top parts are $y_{i,j}^2$. §3.4 gives a BFS degree of regularity of $N+1$.	derived
Q-B: does deeper 2-adicity change Kummer D?	**H2_same at $v_2 = 30$** ($p_2 = 3 \cdot 2^{30}+1$). D and width are equal to p_0 ($v_2=18$) and p_1 ($v_2=27$) at every (m,N), and so are the multiply-adds of p_1 on this build. **$v_2 \geq 40$ was not run.** Both engines require $p < 2^{32}$ (32-bit coefficients), and no prime below 2^{32} has $v_2(p-1) > 30$.	measured / blocked
Q-C: $m=3, N=21$ and $m=4, N=20$?	About 130–150 GB and $2.5–7.4 \cdot 10^{15}$ multiply-adds; about 240–380 GB and $0.85–2 \cdot 10^{16}$. They need a 256 GB or 512 GB host, or a disk-backed B'. Not run.	derived

2. What ran

Shared 4-core, 15 GB container; each run single-threaded, `nice -n 10`, 2.5 GB address-space cap. Two random targets per cell; seed `0x504B4D54`; flags as round 2 (`--engine tower --budget 7200 --max-nnz 2e9`). Primes: torus $T_{24} = 7 \cdot 2^{21}-1$ (14,680,063) and $T_{31} = 61 \cdot 2^{25}-1$ (2,046,820,351); split Dickson at $p_1 = 15 \cdot 2^{27}+1$; deep Kummer at $p_2 = 3 \cdot 2^{30}+1$; positive control (isogeny) at $p_0 = 3 \cdot 2^{18}+1$. The torus builder checks that ζ has exact order 2^t in T, that $u \in T \setminus F_p$ and $u^2 \notin \mu_{2^t}$, that $V \subset F_p$ (zero i-part of $w + w^{-1}$), $|V| = 2^t$, that y^2-4 is a non-residue on V, that no level is ramified, and that $c \neq \pm 2$.

- **Positive control (R0): passed.** Isogeny $D=5$ at $N=4..10$, where the split towers give 4.
- **Exhaustive check:** `pkm_tower_check verify`, 127 finished tower rows, 127 agree.
- **Run status:** no system timed out or ran out of memory. The session's 2-hour task limit stopped the run script during R14 before it wrote a row. R14 was restarted whole and finished. R15 (the second targets at $m=3$, $N=15$) was not run, because the protocol deadline had passed.
- **Build provenance:** R0–R13 used a binary built in the lanes' shared target directory. A rebuild in this lane's own target directory re-measured a subset of every (kind, m, control) series, 144 rows. Every field except wall time was identical (`compare-rows`, 0 differences).

3. Task A: torus against split, measured (D / width / steps)

m, N	torus T_{31}	torus T_{24}	split Dickson p_1	split recorded (Kummer)	torus null T_{31}
2, 2	3/4/2	3/4/2	3/4/2	3	3/4/2
2, 4	4/16/4	4/16/4	4/16/4	4	4/16/4
2, 6	4/56/6	4/56/6	4/56/6	4	4/56/6
2, 8	4/152/8	4/152/8	4/152/8	4	4/152/8
2, 10	4/336/10	4/336/10	4/336/10	4	4/336/10
2, 12	5/1223/13	5/1223/13	5/1223/13	5 (1231)	5/1223/13
2, 14	5/2303/15	5/2303/15	5/2303/15	5 (2304)	5/2303/15
2, 16	5/4786/17	5/4786/17	5/4786/17	5 (4789)	5/4786/17
2, 18	5/8639/20	5/8639/20	5/8639/20	5 (8644)	5/8639/20

2, 20	6/33426/24	6/33426/24	6/33426/24	6 (33446)	6/33426/24
3, 6	5/164/10	5/164/10	5/164/10	5	5/164/10
3, 9	6/1184/16	6/1184/16	6/1184/16	6 (1186)	6/1184/16
3, 12	6/5018/18	6/5018/18	6/5018/18	6 (5019)	6/5066/16
3, 15	7/31378/23 (1)	7/31378/23 (1)	7/31378/23 (1)	7 (31397)	not run
4, 8	6/1080/21	6/1080/21	6/1080/21	6 (1049)	6/952/20
4, 12	7/9451/29	7/9451/29	7/9451/29	7 (9452)	7/8944/33

The T_{24} null equals the T_{31} null in every cell. (1) marks one target. Under rule A1 the torus $m = 2$ line (3 4 4 4 4 5 5 5 5 6) reads inconclusive, as Kummer's does. The shape-matched null gives the torus D everywhere (H1b: no curve-specific effect).

4. Task B: Kummer at $v_2(p-1) = 30$, measured

p_2, $m = 2$, $N = 2..18$: D = 3 4 4 4 4 5 5 5 5, widths 4, 16, 56, 152, 337, 1231, 2304, 4789, 8644. These equal the recorded Kummer values at p_0 and p_1. $m = 3$, $N = 6, 9, 12$: D = 5, 6, 6, widths 164, 1186, 5019, also equal. The null equals the tower in every cell. *Derived:* the Kummer system's coefficients (g, a, b, x_R, and the tower constants 1) do not involve $v_2(p-1)$ beyond the condition $2^t \mid p-1$, so 2-adicity can act only through the prime's size.

5. Task C (derived, not run)

cell	ratio basis	width	peak memory	multiply-adds	time (recorded host)
$m = 3$, $N = 21$	last step	≈ 360k	≈ 146 GB	$2.5 \cdot 10^{15}$	≈ 86 h
$m = 3$, $N = 21$	geometric mean	≈ 475k	≈ 133 GB	$7.4 \cdot 10^{15}$	≈ 138 h
$m = 4$, $N = 20$	last step	≈ 625k	≈ 384 GB	$8.5 \cdot 10^{15}$	≈ 327 h
$m = 4$, $N = 20$	geometric mean	≈ 658k	≈ 243 GB	$2.0 \cdot 10^{16}$	≈ 168 h

The inputs are the recorded Kummer rows at p_1 (requirements README §3.2). The projections assume D does not rise again. If it does, they are too low. The round-6 early exit removes up to 39% of the multiply-adds and does not change memory. The dense reducer block B' holds about 93% of peak memory, so a disk-backed B' is the engine lever. Without one, $m = 3$, $N = 21$ needs about a 256 GB host and $m = 4$, $N = 20$ about a 512 GB host.

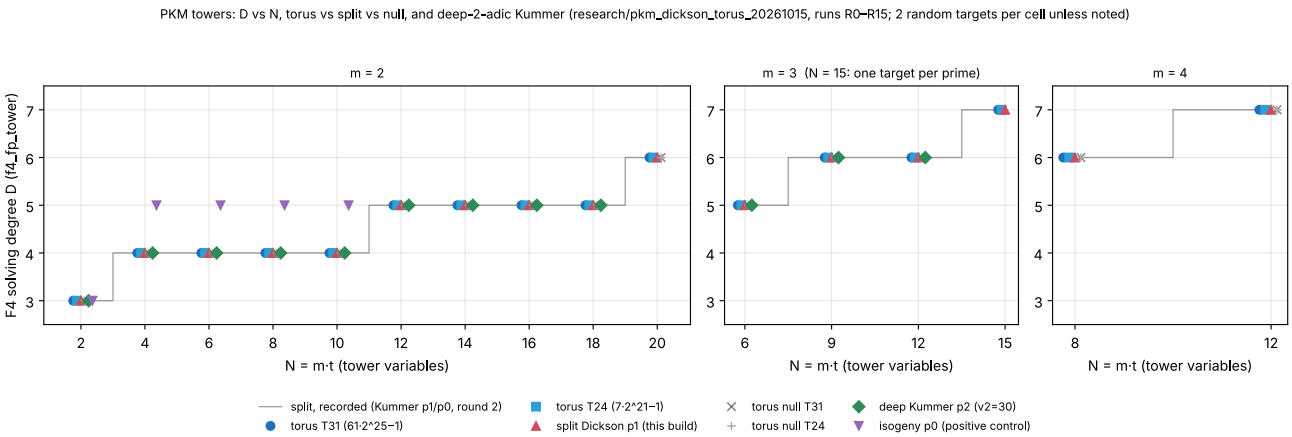


Figure 1: D against N: torus (T_{31}, T_{24}), split Dickson (p_1, this build), torus null, deep-2-adic Kummer (p_2) and the isogeny positive control. The grey line is the recorded split value. Points are offset horizontally; two random targets per cell, one per prime at $m = 3$, $N = 15$. Runs R0–R14.

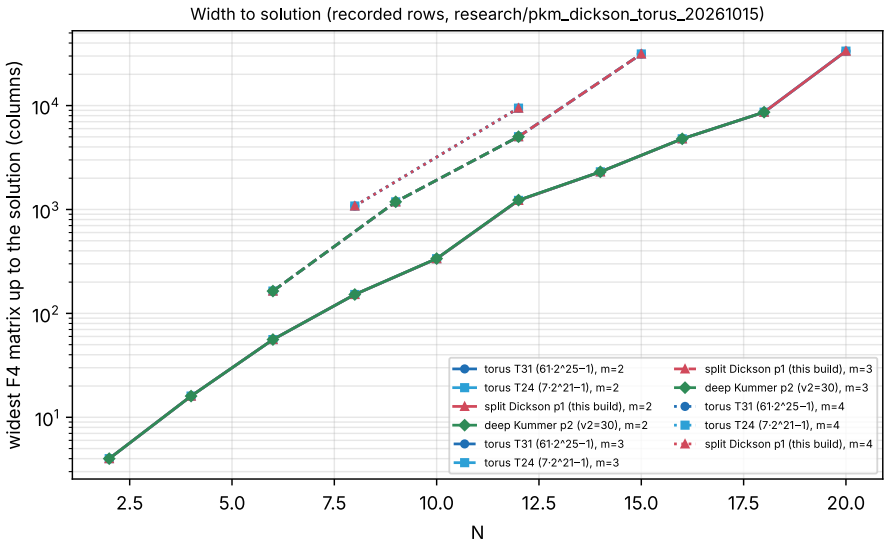


Figure 2: Widest F4 matrix up to the solution (columns, log scale), same runs.

6. Scope

All primes are below 2^{32} and $t \leq 10$, so nothing here is a statement about P-192, P-224 or P-256. Equal D means the torus tower inherits every open question of the split towers (lead 1). It is not evidence that D is bounded. The torus $m = 2$ line rises to 6 at $N = 20$, exactly as Kummer's does.